



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

APAI Lab03: DNN Shrinking and Quantization

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In this Hands-on session:

A first-time user of Pytorch framework will learn how to :

- shrink a NN, by acting on the number of layers, channels, or stride factor
- Quantize a NN down to 2 bits
- Use Netron to visualize a ONNX representation of a CNN

Tasks:

1. Load model's trained weights of LAB1;
2. Reduce network's size under 5 MMAC;
3. Re-train the reduced network and verify network's accuracy;
4. Quantize with QuantLab;
5. Export Onnx and analyze the float32 and quantized models with Netron.

All the details about the tasks are explained in the pdf document attached.



How to deliver the assignment

- Use Virtuale platform to load your file
- update only the .ipynb file, **named as follows:**
LAB1_APAI_yourname.ipynb

Important: the notebook must be pre-run by you. Outputs must be correct and visible when you download it.

LAB3 DEADLINE:
24/10/2025



LAB3 Assignment (due to 25/10/2024 14.30)

Assignment 1 (due 25/11/2021)

Submission status

Submission status	Draft (not submitted)	
Grading status	Not graded	
Due date	Thursday, 25 November 2021, 4:00 PM	
Time remaining	9 days 5 hours	
Last modified	Tuesday, 16 November 2021, 10:56 AM	
File submissions	name of the file  LAB1_APAI_Lorenzo_Lamberti.ipynb 16 November 2021, 10:56 AM	
Submission comments	Comments (0)	

edit submitted files

Edit submission

Remove submission

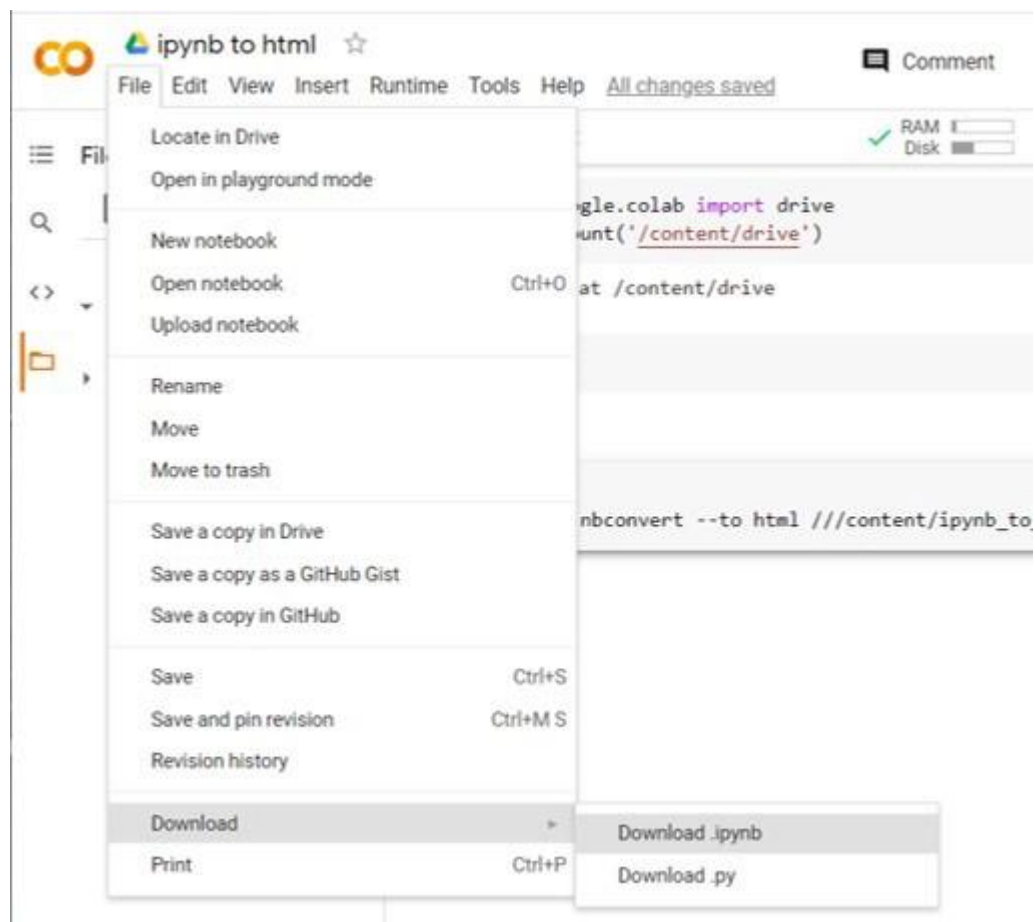
You can still make changes to your submission.

submit

Submit assignment

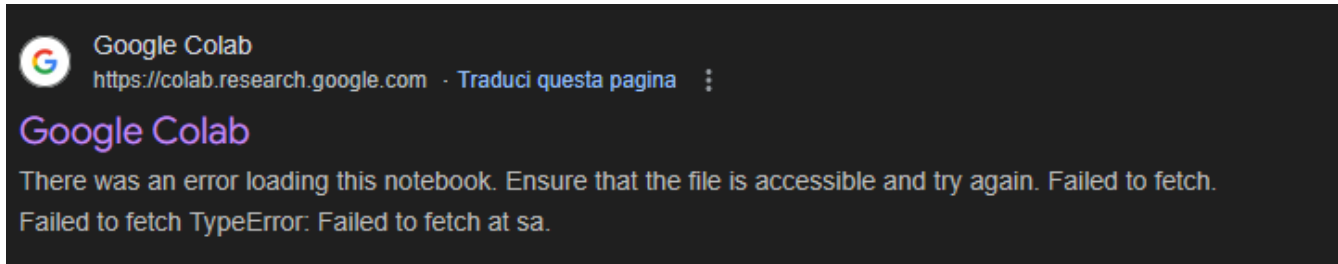


How to download the .ipynb file



Setup

1



Search for and open Google Colab
(<https://colab.research.google.com/>)

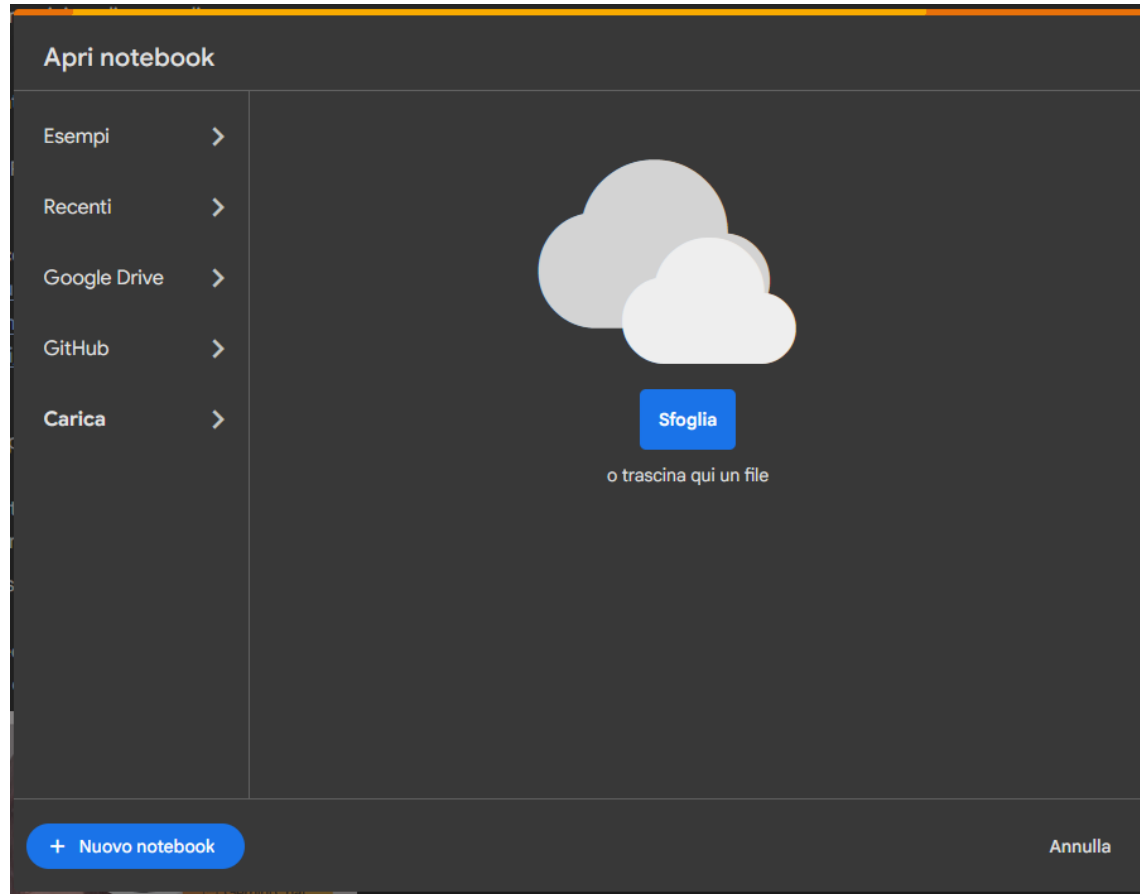
2

Download the .ipynb file from today's repository: <https://github.com/EEESlab/APAI25-LAB03-DNN-Shrinking-and-Quantization>



Setup

3



**Upload the file to Google Colab
(drag and drop or search)**

3

Start working by CAREFULLY reading the setup and assignment text in EVERY block!!



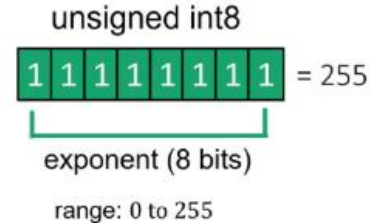
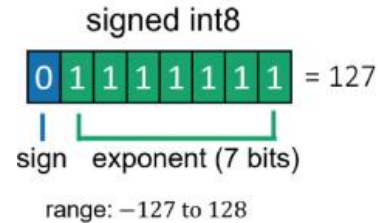
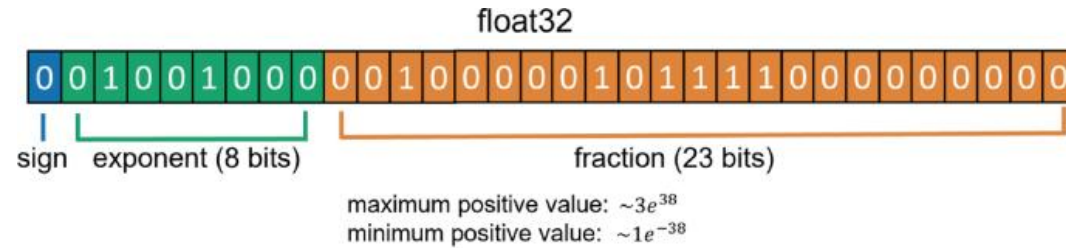


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A bit of theory

Data Types and Quantization

IEEE 754 32-bit Floating Point format



Training time: FP32 for numerical accuracy!
However, 1 element of activation / weights = 4 bytes
(memory hungry for inference!)

Deployment: quantization! Smaller number of bits for a single element (e.g., 8 bits = 1 byte), but less numerical precision!!

Remember Lab1? **Data stored in 8 bits** (instead of 16 or 32) easily **overflow** or cannot represent **values outside range**! One possible solution: **quantization-aware training**!

Sources: https://en.wikipedia.org/wiki/Single-precision_floating-point_format
https://link.springer.com/chapter/10.1007/978-3-031-24538-1_1





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The LAB starts now !

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